

**PROGRAMME: BCA – MACT**

**SEMESTER – V**

**Teaching-Learning & Evaluation Plan (TLEP)**

Course Code: **23BCA5CT02**

Course Title: **Storage Management**

Credits Units: **03**

Contact Hours + Experiential Hours: **45+45**

L-T-P-E: **3-0-0-3**

IA: ESE Weightage – **50:50**

Pass Marks (IA & ESE): **40{ESE-Min:18}**

Aggregate Pass Marks: **40%**

ESE Question Paper Marks: **50**

Special Examination Fees: **NA**

Pre-requisite (if any):

1. Students should know the basics of Computing Devices.
2. Students should know the basics of Networking Architectures for Storage.

**Course Facilitator (s): Dr. Ramkumar Krishnamoorthy, Assistant Professor, School of CS & IT.**

**Programme Outcomes (POs) and Programme Specific Outcomes (PSOs)**

|                                                       |                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| At the end of the Programme, students will be able to |                                                                                                                                                                                                                                                                                                       |
| <b>PO 1</b>                                           | <b>Computational Knowledge:</b> Apply knowledge of computing in fundamentals, specialization, mathematics, and domain knowledge to solve problems and create computing models that represent an abstraction for certain requirements.                                                                 |
| <b>PO 2</b>                                           | <b>Problem Analysis:</b> Identify, formulate, design and solve complex computing problems providing concrete conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.                                                                            |
| <b>PO 3</b>                                           | <b>Develop Solutions:</b> Design, develop and evaluate solutions for computing problems with well-defined components, or processes that meet specified needs.                                                                                                                                         |
| <b>PO 4</b>                                           | <b>Modern Tool / Techniques usage:</b> Select, adapt, and apply appropriate tools, techniques, resources to various computing activities, with an understanding of their limitations.                                                                                                                 |
| <b>PO 5</b>                                           | <b>Professional Ethics:</b> Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practices.                                                                                                                                      |
| <b>PO 6</b>                                           | <b>Life-long learning:</b> Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional.                                                                                                                                         |
| <b>PO 7</b>                                           | <b>Communication Efficiency:</b> Communicate effectively with the computing community, and with society at large, about computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions. |
| <b>PO 8</b>                                           | <b>Societal and Environmental Concern:</b> Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practices.                                           |
| <b>PO 9</b>                                           | <b>Individual and Team work:</b> Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments.                                                                                                                                              |
| <b>PO 10</b>                                          | <b>Innovation and Entrepreneurship:</b> Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.                                                                                           |
| <b>PO11</b>                                           | <b>Conduct Investigations of complex computing problems:</b> Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.                                                   |
| <b>PO12</b>                                           | <b>Project management and finance:</b> Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage                                                                                              |

|               |                                                                                                              |
|---------------|--------------------------------------------------------------------------------------------------------------|
|               | projects and in multidisciplinary environments.                                                              |
| <b>PSO 01</b> | Apply the basic knowledge of Cloud Computing and Android Technology.                                         |
| <b>PSO 02</b> | Analyze and adopt the knowledge on various cloud and mobile service providers.                               |
| <b>PSO 03</b> | Acclimate and apply latest tools, techniques and resources to various cloud and mobile computing activities. |
| <b>PSO 04</b> | Design and develop solutions for real-time problems in Mobile Applications and Cloud Technology.             |

#### Course Objectives:

|             |                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>COB1</b> | Impart the fundamentals of data storage concepts, its demand and Information storage techniques.                      |
| <b>COB2</b> | Instruct different types of hardware required for storage, adapters, connectors, cables and their essential features. |
| <b>COB3</b> | Indoctrinate RAID storage and its techniques.                                                                         |
| <b>COB4</b> | Educate the importance of Backup and Recovery methods.                                                                |
| <b>COB5</b> | Enlighten the Replication and its Consistency features along with Replication Methods.                                |

#### Course Outcomes:

At the end of the course, students will be able to

| Sl. No. | Course Outcome | Description                                                                            | Bloom's Taxonomy Level |
|---------|----------------|----------------------------------------------------------------------------------------|------------------------|
| 1       | CO 1           | Explain the fundamentals of data and information storage                               | L2                     |
| 2       | CO 2           | Use the hardware parts, connectors, SDD, HDD for handling file processing              | L3                     |
| 3       | CO 3           | Interpret RAID to know the functions for understanding the Disk performances           | L3                     |
| 4       | CO 4           | Examine the skills in taking backup and recovery methods for data restoring            | L4                     |
| 5       | CO 5           | Prepare the replications for data preservation and its consistency on the file system. | L5                     |

**CO-PO/PSO Mapping:** (3-Strong Correlation 2- Medium Correlation 1- Low Correlation)

| Course Outcome | Blooms Taxonomy Level | Program Outcomes (PO) |       |       |       |       |       |       |       |       |       |       |       | Program Specific Outcomes(PSO) |               |               |               |
|----------------|-----------------------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------------------|---------------|---------------|---------------|
|                |                       | PO 01                 | PO 02 | PO 03 | PO 04 | PO 05 | PO 06 | PO 07 | PO 08 | PO 09 | PO 10 | PO 11 | PO 12 | PSO 1<br>(L2)                  | PSO 2<br>(L4) | PSO 3<br>(L5) | PSO 4<br>(L4) |
| CO01           | L2                    | 3                     | 2     | -     | 1     | -     | 2     | 1     | -     | 3     | 2     | -     | 2     | 1                              | 2             | 3             | 3             |
| CO02           | L3                    | -                     | 2     | 3     | 3     | -     | 2     | -     | 1     | -     | 2     | 3     | 1     | 3                              | 2             | 1             | 2             |
| CO03           | L3                    | -                     | 3     | 2     | 3     | 1     | -     | 2     | 3     | 2     | -     | 1     | 2     | 1                              | 3             | 2             | 1             |
| CO04           | L4                    | 3                     | 3     | 2     | 3     | -     | 3     | 1     | 2     | -     | 2     | 1     | -     | 3                              | 1             | 2             | 2             |
| CO05           | L5                    | 2                     | 3     | 3     | 2     | -     | 1     | -     | 1     | 1     | 1     | 2     | 2     | 1                              | 3             | 1             | 3             |

### Course Contents:

| Course Contents:             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Module                       | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Contact Hours |
| I                            | <b>Introduction to Information storage and Management:</b><br>Introduction to Information Storage and Management: Information Storage: Data – Types of Data – Information - Storage, Evolution of Storage Technology and Architecture, Data Centre Infrastructure - Core elements- Key Requirements for Data Centre Elements -Managing Storage Infrastructure, Key Challenges in Managing Information, Information Lifecycle - Information Lifecycle Management - ILM Implementation - ILM Benefits                                                                                                                               | 9             |
| II                           | <b>Storage System Environment:</b><br>Components of a Storage System Environment – Host –Connectivity – Storage, Disk Drive Components – Platter – Spindle - Read/Write Head - Actuator Arm Assembly - Controller - Physical Disk Structure -SSD & HDD- Types of hard drive connections - Blade server and Rack server -Zoned Bit Recording - Logical Block Addressing , Disk Drive Performance -I Disk Service Time ,Fundamental Laws Governing Disk Performance , Logical Components of the Host - Operating System -Device Driver -Volume Manager - File System – Application , Application Requirements and Disk Performance. | 9             |
| III                          | <b>RAID and Storage Networking Technologies:</b><br>Implementation of RAID - Software RAID - Hardware RAID -RAID Array Component-Different RAID controllers -RAID Levels -Striping -Mirroring -Parity RAID 0 RAID 1 -Nested RAID -RAID 3 -RAID 4 - RAID 5 --RAID 6 -RAID Comparison -RAID Impact on Disk-Performance - Application IOPS and RAID Configurations -Introduction to Direct Attached Storage – Types of DAS – Introduction to SAN – Components of SAN –FC connectivity – FC topologies – Introduction to NAS – NAS components – NAS Implementation – NAS File sharing                                                 | 9             |
| IV                           | <b>Backup and Recovery:</b><br>Introduction to Business Continuity - Backup Purpose -Disaster Recovery - Operational Backup – Archival, Backup Considerations, Backup Granularity, Recovery Considerations, Backup Methods, Backup Process, Backup and Restore Operations, Backup Topologies - Server less Backup Technologies -Backup to Tape - Physical Tape Library - Backup to Disk - Virtual Tape Library                                                                                                                                                                                                                    | 9             |
| V                            | <b>Replication – Local and Remote:</b><br>Replication – Local and Remote: Source and Target -Uses of Local Replicas, Data Consistency - Consistency of a Replicated File System -Consistency of a Replicated Database, Local Replication Technologies - Host-Based Local Replication -Storage Array-Based Replication, Restore and Restart Considerations - Tracking Changes to Source and Target, Creating Multiple Replicas, Management Interface – Remote Replication Modes                                                                                                                                                    | 9             |
| <b>Text Books (TB):</b>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| TB1                          | EMC Education Services, “Information Storage and Management: Storing, Managing, and Protecting Digital Information”, Wiley Publishing Inc., 1st edition, 2009.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |               |
| <b>Reference Books (RB):</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |
| RB1                          | Robert Spalding, “Storage Networks: The Complete Reference “, Tata McGraw Hill Publication, 2003.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |

**Session-Wise Plan:**

| Abbreviations & Expansions |                                          |                  |                            |
|----------------------------|------------------------------------------|------------------|----------------------------|
| Pedagogy/Activity planned  |                                          | Mode of Delivery |                            |
| P1                         | PPT presentation & Classroom interaction | M1               | Synchronous – PPT          |
| P2                         | Blended Learning with Hands on           | M2               | Asynchronous / synchronous |
| P3                         | Flip Class & Module Quiz                 | M3               | Synchronous - Hands on     |
| A1                         | Activity-1                               | M4               | Synchronous - Discussion   |
| A2                         | Activity-2                               |                  |                            |

| Module   | Session No | Topic Name                                                                            | No. of Session | Readings and References       | Pedagogy/ Activity Planned | CO Mapping | Mode of Delivery | Link to Materials                                                                                                                                                                                 | Done |
|----------|------------|---------------------------------------------------------------------------------------|----------------|-------------------------------|----------------------------|------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>I</b> | 1          | Introduction to Information storage and Management:                                   | 1              | TB-1<br>RB-1<br>VL-1<br>WTL-1 | P1                         | CO1        | M1               | <a href="https://docs.google.com/presentation/u/1/d/1qnLFWzsKueP9cMekFPQLjWbJWf6T6ePq/edit">https://docs.google.com/presentation/u/1/d/1qnLFWzsKueP9cMekFPQLjWbJWf6T6ePq/edit</a>                 |      |
|          | 2          | Information Storage: Data – Types of Data –Information - Storage                      | 1              | TB-1<br>RB-1<br>VL-1<br>WTL-1 | P1                         | CO1        | M1<br>M2         | <a href="https://docs.google.com/presentation/u/1/d/1qnLFWzsKueP9cMekFPQLjWbJWf6T6ePq/edit">https://docs.google.com/presentation/u/1/d/1qnLFWzsKueP9cMekFPQLjWbJWf6T6ePq/edit</a>                 |      |
|          | 3          | Evolution of Storage Technology and Architecture                                      | 1              | TB-1<br>RB-1<br>VL-1<br>WTL-1 | P1                         | CO1        | M1<br>M2         | <a href="https://docs.google.com/presentation/u/1/d/1nnX8M8mLW2NhlFumpCc_jpmDLPGwZ7Dh/edit">https://docs.google.com/presentation/u/1/d/1nnX8M8mLW2NhlFumpCc_jpmDLPGwZ7Dh/edit</a>                 |      |
|          | 4          | Data Centre Infrastructure - Core elements- Key Requirements for Data Centre Elements | 1              | TB-1<br>RB-1<br>VL-1<br>WTL-1 | P1                         | CO1        | M1               | <a href="https://docs.google.com/presentation/d/1Haa200jfXdF8MWqc_WvIQDE_LCe4hdIF/edit">https://docs.google.com/presentation/d/1Haa200jfXdF8MWqc_WvIQDE_LCe4hdIF/edit</a>                         |      |
|          | 5          | Managing Storage Infrastructure, Key Challenges in                                    | 1              | TB-1<br>RB-1<br>VL-1<br>WTL-1 | P1                         | CO1        | M1               | <a href="https://docs.google.com/presentation/d/1C98uKdXv4stvhcbW_Emv0Khqdoq7H-fp/edit?rtfpof=true">https://docs.google.com/presentation/d/1C98uKdXv4stvhcbW_Emv0Khqdoq7H-fp/edit?rtfpof=true</a> |      |

|  |    |                                                                        |   |                               |    |     |    |                                                                                                                                                                                  |  |
|--|----|------------------------------------------------------------------------|---|-------------------------------|----|-----|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  |    | Managing Information                                                   |   |                               |    |     |    |                                                                                                                                                                                  |  |
|  | 6  | Information Lifecycle - Information Lifecycle Management               | 1 | TB-1<br>RB-1<br>VL-1<br>WTL-1 | P1 | CO1 | M1 | <a href="https://docs.google.com/presentation/d/1S8KnlfxDdKjZePPA2yWgITMEHVPXijr-/edit">https://docs.google.com/presentation/d/1S8KnlfxDdKjZePPA2yWgITMEHVPXijr-/edit</a>        |  |
|  | 7  | ILM Implementation -ILM Benefits                                       | 1 | TB-1<br>RB-1<br>VL-1<br>WTL-1 | P1 | CO1 | M1 | <a href="https://docs.google.com/presentation/u/1/d/1sHjCV2Wu8MGtcxOfwBMYnYrjYBUYHmC/edit">https://docs.google.com/presentation/u/1/d/1sHjCV2Wu8MGtcxOfwBMYnYrjYBUYHmC /edit</a> |  |
|  | 8  | <b>Chapter Summary</b>                                                 | 1 | TB-1<br>RB-1<br>VL-1<br>WTL-1 | P1 | CO1 | M1 | Discussion about the Unit 1                                                                                                                                                      |  |
|  | 9  | <b>Quiz</b>                                                            | 1 | TB-1<br>RB-1<br>VL-1<br>WTL-1 | P1 | CO1 | M4 | Sample Questions and Answers discussion                                                                                                                                          |  |
|  | 10 | Storage System Environment: Components of a Storage System Environment | 1 | TB-1<br>RB-1<br>VL-2<br>WTL-2 | P1 | CO2 | M2 |                                                                                                                                                                                  |  |
|  | 11 | Host – Connectivity – Storage, Disk Drive Components –                 | 1 | TB-1<br>RB-1<br>VL-2<br>WTL-2 | P1 | CO2 | M1 |                                                                                                                                                                                  |  |

## II

|    |    |                                                                                                                                               |   |                               |    |     |          |  |  |
|----|----|-----------------------------------------------------------------------------------------------------------------------------------------------|---|-------------------------------|----|-----|----------|--|--|
| II |    | Platter –<br>Spindle -<br>Read/Write<br>Head                                                                                                  |   |                               |    |     |          |  |  |
|    | 12 | Actuator Arm<br>Assembly -<br>Controller -<br>Physical Disk<br>Structure                                                                      | 1 | TB-1<br>RB-1<br>VL-2<br>WTL-2 | P1 | CO2 | M1       |  |  |
|    | 13 | SSD & HDD-<br>Types of hard<br>drive<br>connections                                                                                           | 1 | TB-1<br>RB-1<br>VL-2<br>WTL-2 | P1 | CO2 | M1<br>M2 |  |  |
|    | 14 | Blade server<br>and Rack<br>server -Zoned<br>Bit Recording -<br>Logical Block<br>Addressing                                                   | 1 | TB-1<br>RB-1<br>VL-2<br>WTL-2 | P1 | CO2 | M2       |  |  |
|    | 15 | Disk Drive<br>Performance -1<br>Disk Service<br>Time,<br>Fundamental<br>Laws<br>Governing<br>Disk<br>Performance,<br>Logical<br>Components of | 1 | TB-1<br>RB-1<br>VL-2<br>WTL-2 | P1 | CO2 | M2       |  |  |

|  |    |                                                                                                                              |   |                               |    |     |    |                                         |  |
|--|----|------------------------------------------------------------------------------------------------------------------------------|---|-------------------------------|----|-----|----|-----------------------------------------|--|
|  |    | the Host                                                                                                                     |   |                               |    |     |    |                                         |  |
|  | 16 | Operating System - Device Driver - Volume Manager - File System – Application, Application Requirements and Disk Performance | 1 | TB-1<br>RB-1<br>VL-2<br>WTL-2 | P1 | CO2 | M2 |                                         |  |
|  | 17 | <b>Chapter Summary</b>                                                                                                       | 1 | TB-1<br>RB-1<br>VL-2<br>WTL-2 | P1 | CO2 | M1 | Discussion about the Unit 2             |  |
|  | 18 | <b>Quiz</b>                                                                                                                  | 1 | TB-1<br>RB-1<br>VL-2<br>WTL-2 | P1 | CO2 | M4 | Sample Questions and Answers discussion |  |
|  | 19 | RAID and Storage Networking Technologies: Implementation of RAID - Software RAID - Hardware RAID                             | 1 | TB-1<br>RB-1<br>VL-3<br>WTL-3 | P1 | CO3 | M1 |                                         |  |
|  | 20 | RAID Array                                                                                                                   | 1 | TB-1                          | P1 | CO3 | M1 |                                         |  |

### III

|     |    |                                                                                                        |   |                               |    |     |    |  |  |
|-----|----|--------------------------------------------------------------------------------------------------------|---|-------------------------------|----|-----|----|--|--|
| III |    | Component-Different RAID controllers - RAID Levels                                                     |   | RB-1<br>VL-3<br>WTL-3         |    |     |    |  |  |
|     | 21 | Striping -<br>Mirroring -<br>Parity RAID 0<br>RAID 1 -<br>Nested RAID                                  | 1 | TB-1<br>RB-1<br>VL-3<br>WTL-3 | P1 | CO3 | M1 |  |  |
|     | 22 | RAID 3 -RAID 4 -RAID 5 --<br>RAID 6 -RAID Comparison                                                   | 1 | TB-1<br>RB-1<br>VL-3<br>WTL-3 | P1 | CO3 | M1 |  |  |
|     | 23 | RAID Impact on Disk-Performance -<br>Application IOPS and RAID Configurations                          | 1 | TB-1<br>RB-1<br>VL-3<br>WTL-3 | P1 | CO3 | M1 |  |  |
|     | 24 | Introduction to Direct Attached Storage –<br>Types of DAS - Introduction to SAN –<br>Components of SAN | 1 | TB-1<br>RB-1<br>VL-3<br>WTL-3 | P1 | CO3 | M1 |  |  |

|  |    |                                                                                                                |   |                               |    |     |    |                                         |  |
|--|----|----------------------------------------------------------------------------------------------------------------|---|-------------------------------|----|-----|----|-----------------------------------------|--|
|  | 25 | FC connectivity – FC topologies – Introduction to NAS – NAS components – NAS Implementation – NAS File sharing | 1 | TB-1<br>RB-1<br>VL-3<br>WTL-3 | P1 | CO3 | M1 |                                         |  |
|  | 26 | <b>Chapter Summary</b>                                                                                         | 1 | TB-1<br>RB-1<br>VL-3<br>WTL-3 | P1 | CO3 | M1 | Discussion about the Unit 3             |  |
|  | 27 | <b>Quiz</b>                                                                                                    | 1 | TB-1<br>RB-1<br>VL-3<br>WTL-3 | P1 | CO3 | M4 | Sample Questions and Answers discussion |  |
|  | 28 | Backup and Recovery: Introduction to Business Continuity - Backup Purpose - Disaster Recovery                  | 1 | TB-1<br>RB-1<br>VL-4<br>WTL-4 | P1 | CO4 | M1 |                                         |  |
|  | 29 | Operational Backup – Archival,                                                                                 | 1 | TB-1<br>RB-1<br>VL-4          | P1 | CO4 | M1 |                                         |  |

**IV**

|    |    |                                                                                                            |   |                               |    |     |    |  |  |
|----|----|------------------------------------------------------------------------------------------------------------|---|-------------------------------|----|-----|----|--|--|
| IV |    | Backup Considerations                                                                                      |   | WTL-4                         |    |     |    |  |  |
|    | 30 | Backup Granularity, Recovery Considerations, Backup Methods, Backup Process, Backup and Restore Operations | 1 | TB-1<br>RB-1<br>VL-4<br>WTL-4 | P1 | CO4 | M1 |  |  |
|    | 31 | Backup Topologies - Server less Backup Technologies                                                        | 1 | TB-1<br>RB-1<br>VL-4<br>WTL-4 | P1 | CO4 | M1 |  |  |
|    | 32 | Backup to Tape                                                                                             | 1 | TB-1<br>RB-1<br>VL-4<br>WTL-4 | P1 | CO4 | M1 |  |  |
|    | 33 | Physical Tape Library                                                                                      | 1 | TB-1<br>RB-1<br>VL-4<br>WTL-4 | P1 | CO4 | M1 |  |  |
|    | 34 | Backup to Disk - Virtual Tape Library                                                                      | 1 | TB-1<br>RB-1<br>VL-4<br>WTL-4 | P1 | CO4 | M1 |  |  |

|   |    |                                                                           |   |                               |    |     |    |                                         |  |
|---|----|---------------------------------------------------------------------------|---|-------------------------------|----|-----|----|-----------------------------------------|--|
|   | 35 | Chapter Summary                                                           | 1 | TB-1<br>RB-1<br>VL-4<br>WTL-4 | P1 | CO4 | M1 | Discussion about the Unit 4             |  |
|   | 36 | Quiz                                                                      | 1 | TB-1<br>RB-1<br>VL-4<br>WTL-4 | P1 | CO4 | M4 | Sample Questions and Answers discussion |  |
| V | 37 | Replication – Local and Remote: Source and Target -Uses of Local Replicas | 1 | TB-1<br>RB-1<br>VL-5<br>WTL-4 | P1 | CO5 | M1 |                                         |  |
|   | 38 | Data Consistency - Consistency of a Replicated File System                | 1 | TB-1<br>RB-1<br>VL-5<br>WTL-4 | P1 | CO5 | M1 |                                         |  |
|   | 39 | Consistency of a Replicated Database, Local Replication Technologies      | 1 | TB-1<br>RB-1<br>VL-5<br>WTL-4 | P1 | CO5 | M1 |                                         |  |
|   | 40 | Host-Based Local Replication - Storage Array-Based                        | 1 | TB-1<br>RB-1<br>VL-5<br>WTL-4 | P1 | CO5 | M1 |                                         |  |

|  |    |                                                                   |   |                               |    |     |    |                                         |  |
|--|----|-------------------------------------------------------------------|---|-------------------------------|----|-----|----|-----------------------------------------|--|
|  |    | Replication                                                       |   |                               |    |     |    |                                         |  |
|  | 41 | Restore and Restart Considerations                                | 1 | TB-1<br>RB-1<br>VL-5<br>WTL-4 | P1 | CO5 | M1 |                                         |  |
|  | 42 | Tracking Changes to Source and Target, Creating Multiple Replicas | 1 | TB-1<br>RB-1<br>VL-5<br>WTL-4 | P1 | CO5 | M1 |                                         |  |
|  | 43 | Management Interface – Remote Replication Modes                   | 1 | TB-1<br>RB-1<br>VL-5<br>WTL-4 | P1 | CO5 | M1 |                                         |  |
|  | 44 | <b>Chapter Summary</b>                                            | 1 | TB-1<br>RB-1<br>VL-5<br>WTL-4 | P1 | CO5 | M1 | Discussion about the Unit 5             |  |
|  | 45 | <b>Quiz</b>                                                       | 1 | TB-1<br>RB-1<br>VL-5<br>WTL-4 | P1 | CO5 | M4 | Sample Questions and Answers discussion |  |

**Online Demonstration**

<http://qsm.qsan.net/>

### Video Links: WVL

|      |          |                                                                                                       |
|------|----------|-------------------------------------------------------------------------------------------------------|
| VL-1 | Module-1 | <a href="https://www.youtube.com/watch?v=mUgEgkV16Bw">https://www.youtube.com/watch?v=mUgEgkV16Bw</a> |
| VL-2 | Module-2 | <a href="https://www.youtube.com/watch?v=NgVm2kUmrYo">https://www.youtube.com/watch?v=NgVm2kUmrYo</a> |
| VL-3 | Module-3 | <a href="https://www.youtube.com/watch?v=wTexRObq738">https://www.youtube.com/watch?v=wTexRObq738</a> |
| VL-4 | Module-4 | <a href="https://www.youtube.com/watch?v=hciSIFsbdJs">https://www.youtube.com/watch?v=hciSIFsbdJs</a> |
| VL-5 | Module-5 | <a href="https://www.youtube.com/watch?v=fUrKt-AQYtE">https://www.youtube.com/watch?v=fUrKt-AQYtE</a> |

### Web Text Links: WTL

|       |          |                                                                                                                                                                                                         |
|-------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WTL-1 | Module-1 | <a href="https://www.techopedia.com/definition/30481/storage-management">https://www.techopedia.com/definition/30481/storage-management</a>                                                             |
| WTL-2 | Module-2 | <a href="http://study.com/academy/lesson/magnetic-storage-definition-devices-examples.html">http://study.com/academy/lesson/magnetic-storage-definition-devices-examples.html</a>                       |
| WTL-3 | Module-3 | <a href="https://www.prepressure.com/library/technology/raid">https://www.prepressure.com/library/technology/raid</a>                                                                                   |
| WTL-4 | Module-4 | <a href="https://searchdisasterrecovery.techtarget.com/definition/business-continuity">https://searchdisasterrecovery.techtarget.com/definition/business-continuity</a>                                 |
| WTL-5 | Module-5 | <a href="https://www.computerweekly.com/feature/Remote-replication-Comparing-data-replication-methods">https://www.computerweekly.com/feature/Remote-replication-Comparing-data-replication-methods</a> |

**Blended Learning [P2-Blended Learning with Hands on] :****Interaction - (30 minutes) Teacher Input (20 minutes) Wrap Up (10 minutes)**

| <b>Sl. No</b> | <b>LinkedIn Learning Link</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>      | <a href="https://www.linkedin.com/learning-login/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Fintroduction-to-it-architecture%3Ftrk%3Dshare_ent_url%26shareId%3DOoF0oLAHRGeg4KKG6YyFsg%253D%253D">https://www.linkedin.com/learning-login/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Fintroduction-to-it-architecture%3Ftrk%3Dshare_ent_url%26shareId%3DOoF0oLAHRGeg4KKG6YyFsg%253D%253D</a> |
| <b>2</b>      | <a href="https://www.linkedin.com/learning-login/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Flinux-system-engineer-iscsi-storage-area-">https://www.linkedin.com/learning-login/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Flinux-system-engineer-iscsi-storage-area-</a>                                                                                                                   |
| <b>3</b>      | <a href="https://www.linkedin.com/learninglogin/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Fsystem-administration-backup-and-">https://www.linkedin.com/learninglogin/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Fsystem-administration-backup-and-</a>                                                                                                                                     |
| <b>4</b>      | <a href="https://www.linkedin.com/learning-login/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Fintroduction-to-san-and-nas-">https://www.linkedin.com/learning-login/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Fintroduction-to-san-and-nas-</a>                                                                                                                                             |
| <b>5</b>      | <a href="https://www.linkedin.com/learning-login/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Fcomptia-a-plus-220-1001-cert-prep-4-storage-and-">https://www.linkedin.com/learning-login/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Fcomptia-a-plus-220-1001-cert-prep-4-storage-and-</a>                                                                                                     |
| <b>6</b>      | <a href="https://www.linkedin.com/learning-login/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Flinux-storage-systems">https://www.linkedin.com/learning-login/share?account=92695330&amp;forceAccount=false&amp;redirect=https%3A%2F%2Fwww.linkedin.com%2Flearning%2Flinux-storage-systems</a>                                                                                                                                                           |

## **A1 - Activity-1: MOOC Learning Path on SAN and NAS by LinkedIn Learning (Coursera)**

Students must complete a LinkedIn certification course for a minimum of 15hrs

**MOOC (This Course is Compulsory)**

**Coursera:** <https://www.coursera.org/learn/storage-management---part-1/home/welcome>

### **Rubrics: A1**

| Criteria                  | Poor<br>Below Expectation                                                   | Fair Needs<br>Improvement                                       | Good Meets Expectation                                             | Excellent Exceeds<br>Expectation                                                | Weightage |
|---------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------|
| Certificate<br>Completion | (5-6 marks)<br>Completed less<br>than 15 hrs in 1-2<br>days                 | (7-8 marks)<br>Completed 15<br>hrs or less in 2-3<br>days       | (9 marks) Completed<br>required 15 hrs in less 5-6<br>days         | (10 marks)<br>Completed required<br>15 hrs in 7-8 days                          | 10        |
| Report<br>Submission      | (2 marks) Poor<br>Clarity in lessons<br>learned; poorly<br>organised report | (3 marks)<br>Moderate clarity<br>in lessons learned<br>& report | (4 marks) Good Clarity in<br>lessons learned & organised<br>report | (5 marks)<br>Very Good Clarity in<br>lessons learned & well<br>organised report | 5         |
| Oral<br>Presentation      | (2 marks) Poor<br>language, voice,<br>& Content                             | (3 marks)<br>Moderate<br>language, voice,<br>& Content          | (4 marks) Good language,<br>voice, & Content                       | (5 marks)<br>Very Good language,<br>voice, & Content                            | 5         |
| Question-<br>Answer       | (2 marks) Answers<br>less than 3 questions<br>correctly                     | (3 marks)<br>Answers 3<br>questions<br>correctly                | (4 marks)<br>Answers 4 questions<br>correctly                      | (5 marks)<br>Answers all 5 questions<br>correctly                               | 5         |

## A2 - Activity-2: Mini Project/Case Study Report-2

Students must show their project report after collecting from the Industrial Study

### Rubrics: A2

| Criteria                              | Excellent (4)                                                 | Good (3)                                    | Fair (2)                              | Needs Improvement (1)              |
|---------------------------------------|---------------------------------------------------------------|---------------------------------------------|---------------------------------------|------------------------------------|
| <b>Preparation &amp; Research</b>     | Demonstrated strong prior knowledge; asked relevant questions | Basic understanding; asked some questions   | Limited research; few questions asked | No evidence of preparation         |
| <b>Participation &amp; Engagement</b> | Actively engaged with speakers and staff; took notes          | Showed interest; participated in discussion | Minimal interaction or note-taking    | Passive or distracted during visit |
| <b>Observation Skills</b>             | Noted key aspects of storage systems, layout, technologies    | Identified most storage components          | Missed several important observations | Little attention to details        |
| <b>Understanding of Concepts</b>      | Clearly connected visit to storage management theories        | Made good connections with classroom topics | Partial understanding shown           | Struggled to relate concepts       |

|                                        |                                                                |                                      |                                  |                                  |
|----------------------------------------|----------------------------------------------------------------|--------------------------------------|----------------------------------|----------------------------------|
| <b>Reflection &amp; Report Quality</b> | Insightful, well-organized report with real-world applications | Complete report with good analysis   | Report lacks depth and structure | Poorly written or missing report |
| <b>Teamwork &amp; Collaboration</b>    | Worked well with peers; contributed to group activities        | Cooperated with team reasonably well | Some effort to collaborate       | Lacked teamwork or disruptive    |
| <b>Professionalism &amp; Conduct</b>   | Courteous, punctual, appropriately dressed                     | Mostly professional behavior         | Minor behavioral issues          | Unprofessional or disrespectful  |

| <b>Criteria</b>                        | <b>Description</b>                                             | <b>Weightage</b> | <b>Marks Range</b> |
|----------------------------------------|----------------------------------------------------------------|------------------|--------------------|
| <b>Preparation &amp; Research</b>      | Pre-visit study, familiarity with storage systems              | 10%              | 1.5                |
| <b>Participation &amp; Engagement</b>  | Interaction with experts, note-taking, responsiveness          | 20%              | 3.0                |
| <b>Observation Skills</b>              | Identifying technologies, layout, processes during visit       | 13%              | 2.0                |
| <b>Understanding of Concepts</b>       | Connecting observations to theoretical knowledge               | 20%              | 3.0                |
| <b>Reflection &amp; Report Quality</b> | Insightful analysis, report structure, application to learning | 20%              | 3.0                |
| <b>Professionalism &amp; Conduct</b>   | Discipline, behavior, punctuality, dress code                  | 13%              | 2.0                |
| <b>TOTAL</b>                           |                                                                | <b>100%</b>      | <b>15.0</b>        |

**Assessment Scheme: IA: ESE - 50:50**

| <b>Sl. No.</b> | <b>Assessment Instrument</b>               | <b>Formative/ Summative</b> | <b>Frequency</b> | <b>Weightage in Marks</b> | <b>CO</b>                |
|----------------|--------------------------------------------|-----------------------------|------------------|---------------------------|--------------------------|
| 1              | Activity-1 – LinkedIn Certification Course | Formative                   | 1                | <b>15</b>                 | CO 1, CO2, CO3, CO4, CO5 |
| 2              | Activity-2 - MiniProject-2                 | Formative                   | 1                | <b>15</b>                 | CO 1, CO2, CO3, CO4, CO5 |
| 3              | Attendance                                 | Formative                   | -                | <b>5</b>                  | -                        |
| 4              | Internal Exam 1                            | Formative                   | 1                | <b>7.5</b>                | CO1, CO2, CO3            |
| 5              | Internal Exam 2                            | Formative                   | 1                | <b>7.5</b>                | CO3, CO4, CO5            |
| 6              | University Exam                            | Summative                   | 1                | <b>50</b>                 | CO1, CO2, CO3, CO4, CO5  |
|                | <b>Total</b>                               |                             |                  | <b>100</b>                |                          |